

Polynomial Metric Entropy of the Monotone Spherical Chamber

Solution packet for arXiv:1210.0139

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Status. Partial result likely valid, subject to human review. The theorem determines the sharp order in the ambient dimension at every fixed chordal accuracy $0 < \varepsilon \leq 1/5$: the optimal covering number is polynomial in M , whereas the source construction is quasi-polynomial.

1 Source question

Fickus, Jasper, Mixon, and Peterson define

$$\mathbb{S}_{\text{nn}}^{M-1} = \{x \in \mathbb{R}^M : 0 \leq x_1 \leq \cdots \leq x_M, \|x\|_2 = 1\}$$

and use the chordal distance

$$\rho(x, y) = \sqrt{1 - |\langle x, y \rangle|^2}.$$

Their signed-permutation reduction makes an efficient net for this chamber the computational bottleneck in their NERF estimates. In the conclusions they ask, in effect, how many nonnegative monotone unit vectors are needed to approximate all such vectors in chordal distance [1, p. 16].

Let $\mathcal{N}_M(\varepsilon)$ denote the least cardinality of an ε -net for $\mathbb{S}_{\text{nn}}^{M-1}$ in ρ . The source constructs, for fixed ε , a net of size $\exp(O_\varepsilon((\log M)^2))$. The result below improves the dimension dependence to a power of M , and shows that a power is necessary at sufficiently fine fixed accuracy.

2 Proof intuition

The upper bound uses monotonicity in the coordinate index rather than quantization in the value axis. Reverse the coordinates and divide them into geometrically growing consecutive blocks. On a block beginning at a , use at most ηa coordinates. Replacing a decreasing vector by its value at the right endpoint of each block loses at most an η fraction of its squared mass. Thus every chamber point is close to a fixed block-constant subspace of dimension $O(\eta^{-1} \log M)$. A standard net in that subspace gives the upper bound after taking $\eta \asymp \varepsilon^2$.

For the lower bound, reverse the idea. Use $d \asymp \log M$ blocks whose sizes decrease by a factor 16. Give every block nearly equal *energy*, and perturb those energies by balanced signs from a binary code. The factor 16 leaves enough separation between adjacent block heights to preserve monotonicity under every perturbation. Codewords far apart in Hamming distance then give chamber points a fixed positive chordal distance apart.

meaning that for any K the corresponding optimal NERF bounds satisfy $K - (N - \frac{N}{M}) \leq \alpha_K \leq \beta_K \leq \frac{N}{M}$. In particular, it was already apparent that the previously discussed group frame of $N = 560$ vectors in $\mathbb{R}^8$ is robust up to $\frac{N}{M} = 70$ erasures, meaning we could already safely take K as small as 491. However, the results of this paper allow us to say much more: applying Theorem 6 in the case where $\varepsilon = \frac{1}{5}$ tells us K can actually be as small as 399; applying this same result with a smaller ε may very well yield even smaller acceptable values of K .

Conclusions and future work

For many decades, numerical linear algebraists have intensely studied the problem of estimating the extreme eigenvalues of a self-adjoint positive semidefinite matrix. As a result, we have many excellent algorithms—power methods, QR iterations, etc.—for quickly and accurately computing these eigenvalues for even large matrices. However, these algorithms cannot be directly applied to many emerging compressed sensing and frame theory problems, since these problems involve estimating the singular values of all submatrices of a given size, leading to a combinatorial nightmare. What we need are new, computationally tractable algorithms that allow us to estimate the singular values of all of these submatrices simultaneously.

We believe the techniques of this paper are a step in that direction. However, as seen in the last section, these techniques are still computationally expensive. Much of this is the fault of our ε -net for $\mathbb{S}_{nn}^{M-1}$ which, despite having far fewer elements than an ε -net for the entire sphere $\mathbb{S}^{M-1}$, still contains an enormous number of points for even small choices of M . As such, with regards to taking the results of this paper forward, the most critical problem that needs to be addressed is that of designing more efficient ε -nets for $\mathbb{S}_{nn}^{M-1}$. Indeed, this seems to be a fundamental problem in analysis: how many nonnegative, nonincreasing unit norm functions do I need to well-approximate every such function, where the quality of the approximation is measured in terms of chordal distance? Of course, many other important questions arise. Can the optimal NERF bound estimates of Theorems 1 and 2 be improved? Is there

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Figure 1: The open-ended covering question in the source paper, page 16.

3 The metric-entropy theorem

Theorem 1. *There are universal constants $c > 0$ and $M_0 \geq 2$ such that, for every $M \geq M_0$ and $0 < \varepsilon \leq 1/5$,*

$$M^c \leq \mathcal{N}_M(\varepsilon) \leq \left(1 + \frac{4}{\varepsilon}\right)^{32\varepsilon^{-2}(1+\log M)}. \quad (1)$$

Consequently, for each fixed $0 < \varepsilon \leq 1/5$,

$$\log \mathcal{N}_M(\varepsilon) = \Theta_\varepsilon(\log M) \quad (M \rightarrow \infty).$$

The upper bound remains valid for every $0 < \varepsilon < 1$.

Reversing the order of the coordinates is an isometry, so in the upper-bound proof we work with

$$T_M = \{x \in \mathbb{R}^M : x_1 \geq \cdots \geq x_M \geq 0, \|x\|_2 = 1\}.$$

For points in either chamber the inner product is nonnegative, and

$$\rho(x, y)^2 = (1 - \langle x, y \rangle)(1 + \langle x, y \rangle) \leq 2(1 - \langle x, y \rangle) = \|x - y\|_2^2. \quad (2)$$

Lemma 1 (geometric-block compression). *For every $0 < \eta \leq 1$ there is a partition of $\{1, \dots, M\}$ into d consecutive blocks such that*

$$d \leq 4\eta^{-1}(1 + \log M), \quad (3)$$

and the orthogonal projection P onto the subspace of vectors constant on each block satisfies

$$\|x - Px\|_2^2 \leq \eta \quad (x \in T_M). \quad (4)$$

Moreover, Px is nonnegative and nonincreasing.

Proof. Make every index $a < 2/\eta$ a singleton. Thereafter, whenever the next block starts at a , give it length

$$\ell_a = \min\{\lfloor \eta a \rfloor, M - a + 1\}.$$

For a nonterminal nonsingleton block, $\eta a/2 \leq \ell_a \leq \eta a$. Its successor therefore begins at least at $(1 + \eta/2)a$. Since $\log(1 + \eta/2) \geq \eta/4$, counting the initial singletons and subsequent geometric growth gives (3).

Fix $x \in T_M$. On a block $B = [a, b]$, define $q_i = x_b$ for $i \in B$. Then q is block-constant, and

$$\sum_{i \in B} (x_i - x_b)^2 \leq \sum_{i \in B} (x_i^2 - x_b^2). \quad (5)$$

Put $z_i = x_i^2$. For any $t \geq 0$, the set $\{i : z_i > t\}$ is an initial interval $\{1, \dots, k(t)\}$. In the layer-cake representation of the right side of (5), complete blocks cancel. If $k(t)$ lies in a nonsingleton block $[a, b]$, the uncanceled number of indices is at most $|B| \leq \eta a \leq \eta k(t)$; it is zero in an initial singleton. Hence

$$\|x - q\|_2^2 \leq \int_0^\infty \eta k(t) dt = \eta \sum_{i=1}^M z_i = \eta.$$

Orthogonal projection is at least as accurate as q , proving (4). Finally, Px consists of the consecutive block averages of x , so it remains nonnegative and nonincreasing. $\square$

Proof of the upper bound in Theorem 1. Take $\eta = \varepsilon^2/8$ and the subspace V from Lemma 1. If $e = \|x - Px\|_2$, orthogonality gives $\|Px\|_2 = \sqrt{1 - e^2}$. Thus, for $u = Px/\|Px\|_2 \in T_M \cap V$,

$$\|x - u\|_2^2 = 2(1 - \|Px\|_2) \leq 2e^2 \leq 2\eta = \varepsilon^2/4. \quad (6)$$

Take a maximal Euclidean $\varepsilon/2$ -separated subset A of $T_M \cap V$. It is an $\varepsilon/2$ -net of that set. The usual disjoint-ball argument in the d -dimensional space V gives

$$|A| \leq (1 + 4/\varepsilon)^d.$$

Equation (6) and the triangle inequality show that A is an Euclidean ε -net of T_M , and (2) makes it a chordal ε -net as well. Since Lemma 1 gives $d \leq 32\varepsilon^{-2}(1 + \log M)$, the upper estimate follows. $\square$

Proof of the lower bound in Theorem 1. Let d be the largest even integer for which

$$m_d := 1 + 16 + \dots + 16^{d-1} = \frac{16^d - 1}{15} \leq M.$$

Thus $d \geq c_1 \log M$ for all sufficiently large M . After $M - m_d$ initial zero coordinates, split the remaining coordinates into consecutive blocks $B_1, \dots, B_d$ of sizes $|B_j| = 16^{d-j}$.

We use a family $\mathcal{C} \subseteq \{-1, 1\}^d$ of balanced words ($\sum_j \sigma_j = 0$) with pairwise Hamming distance at least $d/4$ and $|\mathcal{C}| \geq \exp(c_2 d)$. For completeness, such a family follows by a greedy selection from the $\binom{d}{d/2}$ balanced words. A Hamming ball of radius less than $d/4$ contains at most

$$\sum_{r \leq d/8} \binom{d/2}{r}^2 \leq (d+1)2^{dH_2(1/4)}$$

balanced words, while $\binom{d}{d/2} \geq 2^d/(d+1)$ and $H_2(1/4) < 1$.

For $\sigma \in \mathcal{C}$, define x^σ to be constant on B_j with value

$$x_i^\sigma = \frac{1 + \sigma_j/2}{\sqrt{(5/4)d|B_j|}} \quad (i \in B_j). \quad (7)$$

Balancedness gives $\|x^\sigma\|_2 = 1$. Adjacent block heights obey

$$\frac{x^\sigma|_{B_{j+1}}}{x^\sigma|_{B_j}} = 4 \frac{1 + \sigma_{j+1}/2}{1 + \sigma_j/2} \geq \frac{4}{3},$$

so every x^σ belongs to the nonnegative nondecreasing chamber.

If h is the Hamming distance between σ and τ , direct calculation using balancedness gives

$$\langle x^\sigma, x^\tau \rangle = 1 - \frac{2(1/2)^2}{1 + (1/2)^2} \frac{h}{d} = 1 - \frac{2}{5} \frac{h}{d}.$$

For $h \geq d/4$, this is at most $9/10$, and hence

$$\rho(x^\sigma, x^\tau) \geq \sqrt{1 - (9/10)^2} = \frac{\sqrt{19}}{10} > \frac{2}{5}.$$

When $\varepsilon \leq 1/5$, an ε -ball contains at most one member of this packing. Therefore

$$\mathcal{N}_M(\varepsilon) \geq |\mathcal{C}| \geq \exp(c_2 d) \geq M^c,$$

after decreasing the universal constant $c > 0$ and increasing M_0 . □

4 Verification and limitations

The argument is elementary. The accompanying script `code/verify_entropy_bounds.py` checks the geometric-block estimate on random and adversarial monotone vectors, verifies the dimension count, and enumerates finite balanced-code packings to check normalization, monotonicity, and the chordal separation constant. These computations are sanity checks, not ingredients of the proof. It passed with 300 random trials at $(M, \varepsilon, d) = (4096, 0.2, 8)$ and 200 trials at $(7777, 0.125, 10)$. The commands were

```
python code/verify_entropy_bounds.py --dimension 4096 --epsilon 0.2 \
--trials 300 --packing-blocks 8
python code/verify_entropy_bounds.py --dimension 7777 --epsilon 0.125 \
--trials 200 --packing-blocks 10
```

In the first run, the minimum enumerated packing distance was exactly $\sqrt{19}/10$, the constant used in the proof.

The theorem settles the dimension dependence of the optimal covering number at every fixed $0 < \varepsilon \leq 1/5$. It does not determine the sharp dependence of the exponent on ε , and the maximal-net step is an existence construction rather than a practically optimized enumeration. Accordingly, this packet is classified as a substantial partial result rather than a full answer to every algorithmic interpretation of the source’s broad question.

A bounded novelty search used the run indexes, arXiv:1210.0139, the exact source phrases “nonnegative, nonincreasing unit norm” and “more efficient ε -nets,” and combinations of “metric entropy,” “monotone cone,” “isotonic cone,” and “chordal distance.” No later paper directly settling this chamber covering question was found. This is a bounded search, not an exhaustive priority claim.

Human-review recommendation. Check the layer-cake cancellation in Lemma 1, the normalization estimate (6), and the constant-weight-code packing. These are the only substantive steps.

References

- [1] M. Fickus, J. Jasper, D. G. Mixon, and J. Peterson, *Group-theoretic constructions of erasure-robust frames*, Linear Algebra Appl. 479 (2015), 131–154; arXiv:1210.0139.